Internship Project Report

Project Title: Civil Engineering Insight Studio

Submitted in partial fulfilment of the requirements for the

Google Cloud Generative AI

conducted by

SmartBridge

Submitted by

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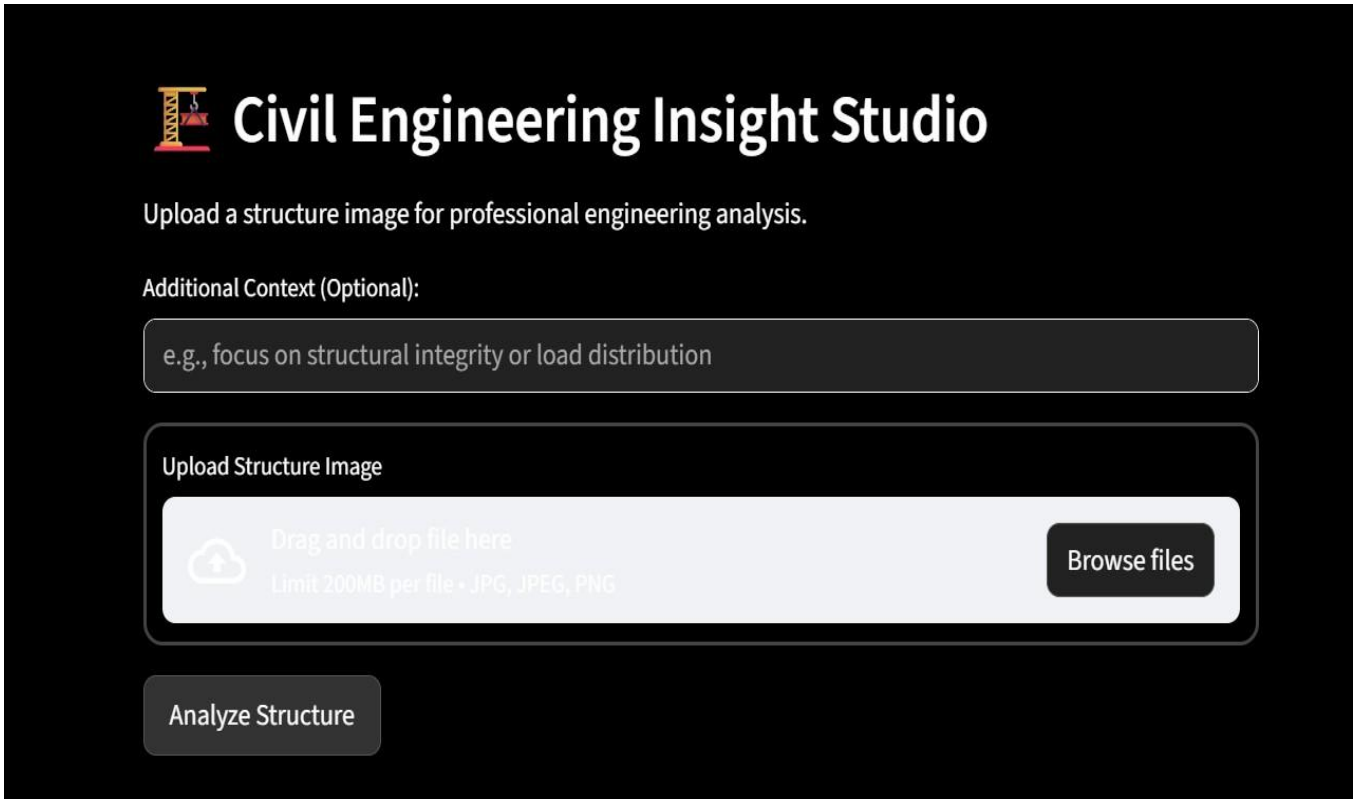


1. INTRODUCTION

1.1 Project Overview

Civil Engineering Insight Studio is an AI-powered web-based application developed to analyse images of civil engineering structures and automatically generate comprehensive structural descriptions. The system leverages advanced computer vision and artificial intelligence technologies to identify structural components such as beams, columns, slabs, foundations, bridges, pavements, and construction materials.

The application processes uploaded images and produces structured technical reports that include:

- Type of structure
- Structural components identified
- Materials used (concrete, steel, composite, etc.)
- Construction methodology (RCC, prestressed, steel frame, etc.)
- Load-bearing elements
- Visible design features

1.2 Purpose

The primary purpose of this project is to automate structural analysis documentation using artificial intelligence and computer vision technologies. Traditional documentation of structural elements is time-consuming and often inconsistent across different engineers or organizations. This system aims to:

- Reduce manual documentation effort
- Improve accuracy and consistency in structural reporting
- Provide quick preliminary structural assessments
- Assist engineers during site inspections
- Support academic learning for civil engineering students

2. IDEATION PHASE

2.1 Problem Statement

Manual structural documentation from images is time consuming and subjective. Engineers require an AI-based system to automatically analyse structures and generate accurate descriptions.

2.2 Empathy Map

User Thinks:

- I need a fast and accurate structural description.

User Feels:

- Overwhelmed by manual reporting tasks.

User Says:

- Preparing structural documentation takes too much time.

User Does:

- Manually inspects site images and writes reports.

2.3 Final Idea

- AI-Based Civil Engineering Structure Description Generator using Google Gemini Vision Model

- The final idea of this project is to develop an AI-Based Civil Engineering Structure Description Generator that utilizes

the Google Gemini Vision Model for intelligent image analysis and automated structural documentation.

3. REQUIREMENT ANALYSIS

Requirement Analysis is a crucial phase in system development that identifies and documents the functional and non-functional requirements of the proposed system. For Civil Engineering Insight Studio, this section outlines what the system must do and the constraints under which it must operate.

3.1 Functional Requirements

1. Accept image uploads (JPG, PNG, JPEG)
2. Accept user input prompts
3. Generate structural description using AI
4. Display structured output

3.2 Non-Functional Requirements

1. Fast response time
2. Secure API management
3. User-friendly interface

3.3 Technology Stack

1.Programming Language: Python

2. Web Framework: Streamlet

3.AI Model: Google Gemini Pro Vision

4. Environment Management: python-dotenv

3.4 Data Flow Diagram (DFD)

Input → Processing → Output

User Inputs → Gemini AI Processing → Generated Itinerary

3.3 Technology Stack

Programming Language – Python

Python is used as the core programming language for developing the system. It is chosen because of its simplicity, readability, and strong support for artificial intelligence, data processing, and API integration.

Web Framework – stream lit

stream lit is used to build the user interface of the application. It allows developers to quickly create interactive web applications using simple Python code without requiring knowledge of HTML, CSS, or JavaScript.

AI Model – Google Gemini 1.5 Flash

Google Gemini 1.5 Flash is used as the core artificial intelligence model in the system. It provides fast and accurate responses for tasks such as text understanding, content generation, and analysis, making it suitable for real-time AI applications. **API Integration – Google Generative AI API**

The Google Generative AI API is used to connect the application with the Gemini AI model. It enables seamless communication between the system and the AI service **Environment Management – python-dotenv** python-dotenv is used to manage environment variables securely. It helps store sensitive data such as API keys and configuration settings in a separate file,

4. PROJECT DESIGN

4.1 Problem–Solution Fit

Civil Engineering Insight Studio effectively solves the problem of manual structural documentation and preliminary visual analysis in civil engineering projects.

4.2 Proposed Solution

A web-based AI application where users upload images of civil engineering structures.

User Input

• Structural image (building, bridge, slab, column, etc.)
• Optional description (if needed)

System Processing

• Analyse the uploaded image
• Detect structural components
• Identify materials used
• Recognize construction type

System Output

The system generates:

• Type of structure
• Identified components (beams, columns, slabs, etc.)
• Materials used
• Construction methodology (RCC, steel, composite, etc.)
• Engineering observations
• Notable design features

The output is presented in a structured, professional format suitable for documentation, academic projects, and preliminary inspections.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning Ideation – 1 Week
In this phase, the project idea was planned and defined. Requirements were gathered, objectives were set, and the overall system design and workflow were outlined.

Development – 2 Weeks
During this phase, the actual implementation of the system was carried out. The application was developed using Python, Streamlit, and AI integration, and core functionalities were built.

Testing – 1 Week
In the testing phase, the system was checked for errors, bugs, and performance issues. Different test cases were executed to ensure the application works correctly and reliably.

Documentation – 1 Week
This phase involved preparing project documentation, including reports, technical details, user guides, and explanations of the system's features and workflow.

6. FUNCTIONAL AND PERFORMANCE TESTING

Performance Testing

1. System responds within seconds
2. Handles multiple inputs efficiently
3. Accurate AI-generated itineraries

Travel.py Code:

```python
import streamlit as st
import os
from google import genai
from PIL import Image
from dotenv import load_dotenv

st.set_page_config(
    page_title="Civil Engineering Insight Studio",
    layout="centered"
)

theme_option = st.sidebar.radio(
    "Select Theme",
    ["Dark", "Light"],
    index=0
)

if theme_option == "Dark":
    st.markdown(
        """
        <style>
            /* Main background */
            .stApp { background-color: #000000; }
            html, body, [class*="css"] { color: #FFFFFF !important; }
            h1,h2,h3,h4,h5,h6,p,label,div,span { color: #FFFFFF !important; }

            /* Sidebar */
            section[data-testid="stSidebar"] { background-color: #111111 !important; }
            section[data-testid="stSidebar"] * { color: #FFFFFF !important; }
```

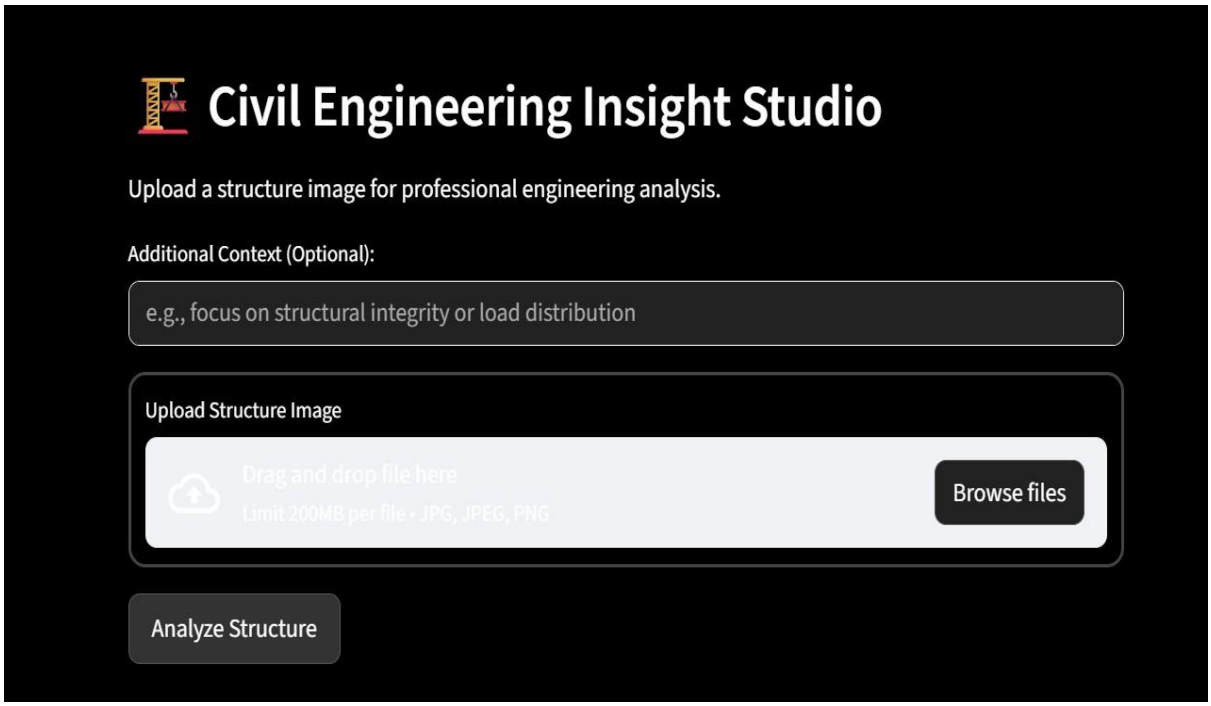
Requirements.txt:

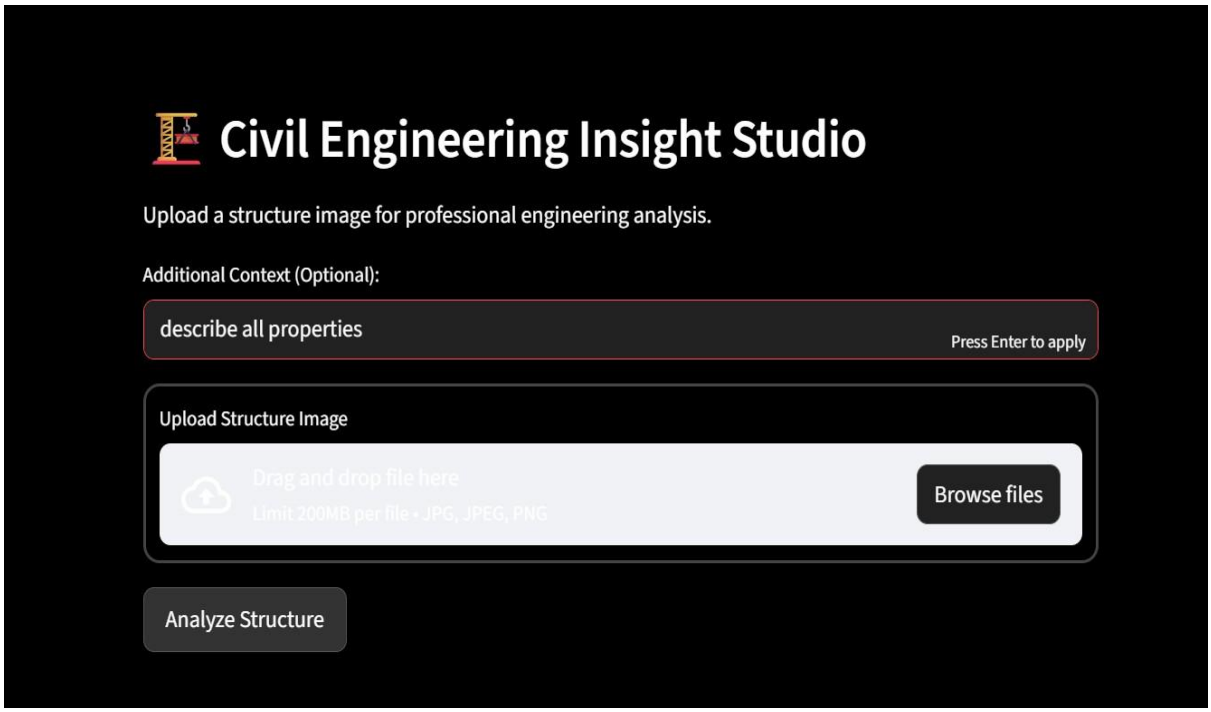
```
streamlit
google-genai
python-dotenv
Pillow
```

Running the Application:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   streamlit

PS D:\GENAI> streamlit run travel.py

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://192.168.10.46:8501
```

User Interface:

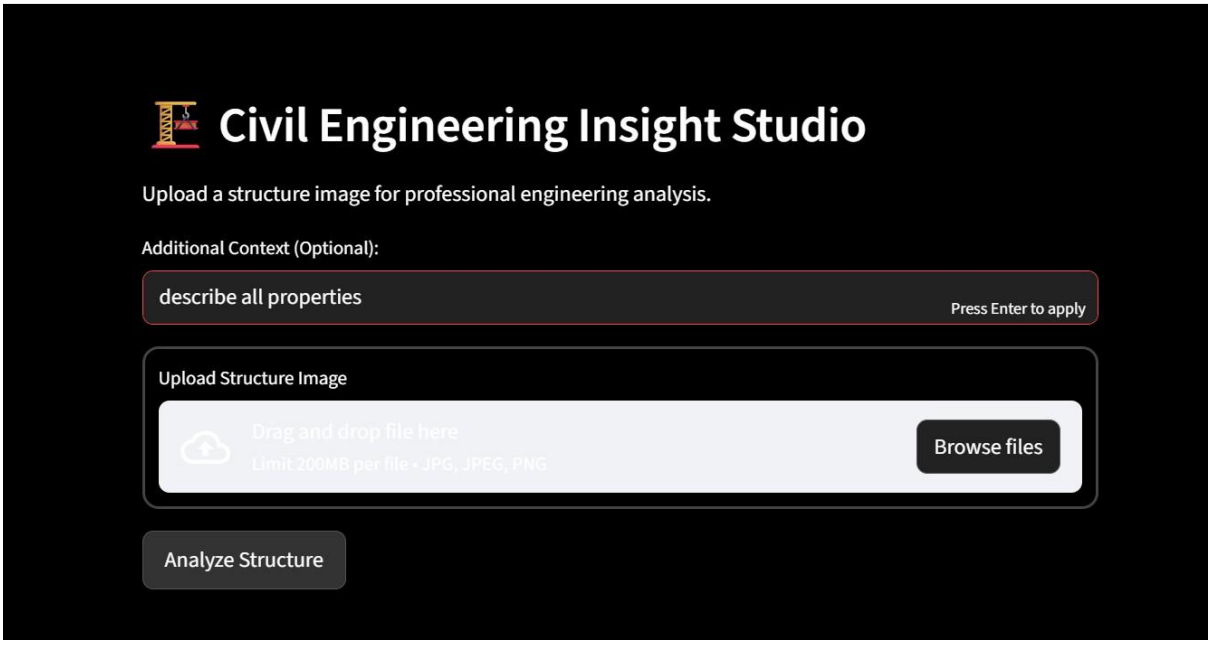


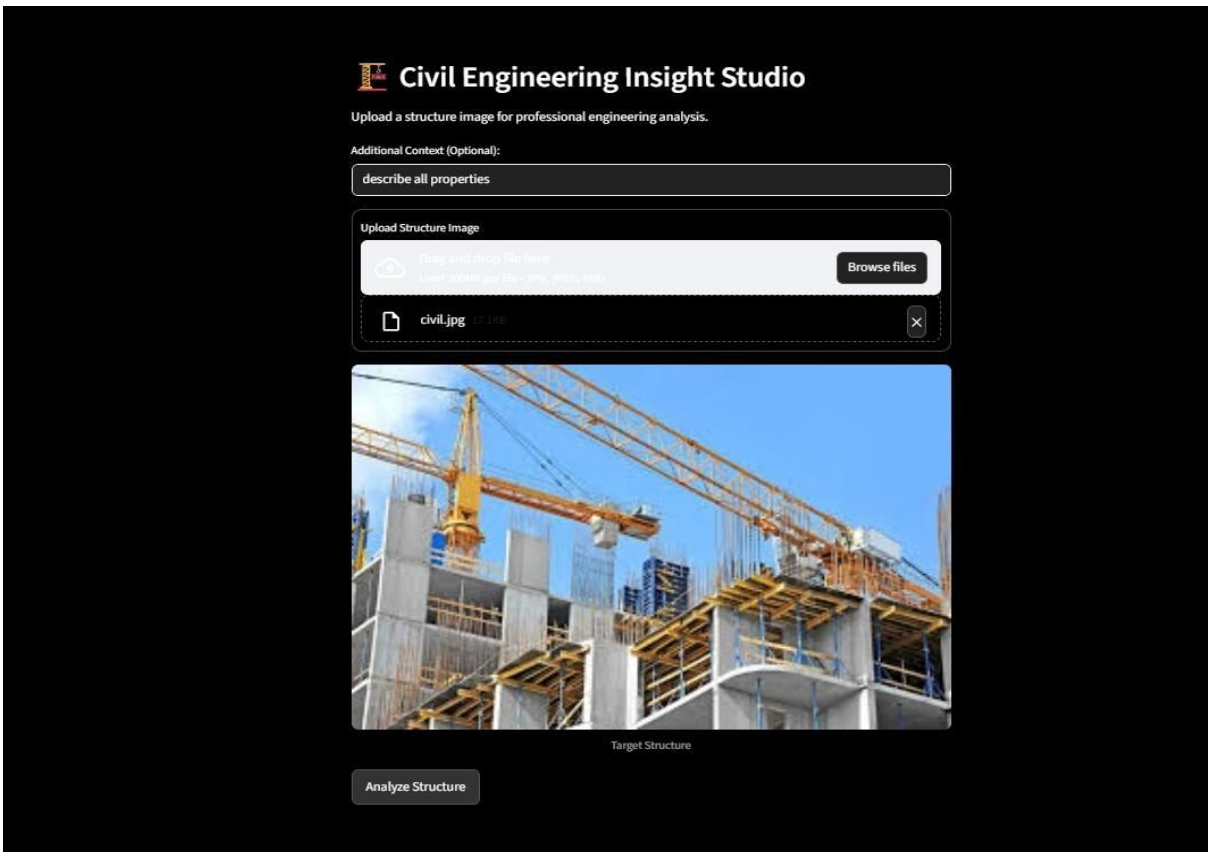


7. RESULTS

7.1 Output Screenshots Include screenshots of:

User input form:





Generated itinerary:

📋 Engineering Report

Professional Engineering Report: Analysis of Multi-Story Reinforced Concrete Structure Under Construction

Date: October 26, 2023 **Subject:** Structural Analysis of Unnamed Construction Project (Image-Based)
Prepared By: Expert Civil Engineer

This report provides a professional engineering analysis of the multi-story structure currently under construction, based solely on the provided photographic evidence.

1. Type of Structure

The structure observed is a **multi-story framed building**, predominantly utilizing a **cast-in-place reinforced concrete system**. Given the repetitive floor plates and apparent grid of columns and beams, it is likely intended for **residential or commercial (office) purposes**. The scale and the use of a tower crane suggest it will be a mid-to-high-rise building.

The primary structural system consists of:

- **Reinforced Concrete Columns:** Vertical load-bearing elements.
- **Reinforced Concrete Beams:** Horizontal load-bearing elements supporting slabs and transferring loads to columns.
- **Reinforced Concrete Slabs:** Horizontal elements forming the floors and roofs.
- **Reinforced Concrete Shear Walls (potential):** While not explicitly visible as full-height shear walls, certain thicker wall elements or solid core sections (e.g., for elevators/stairs) may act as shear walls to resist lateral loads (wind, seismic).

2. Materials Likely Used

Based on visual inspection, the following materials are predominantly employed:

- **Reinforced Concrete:** The primary structural material for columns, beams, and floor slabs. This involves a composite of:
  - **Portland Cement Concrete:** A mixture of cement, aggregates (sand, gravel), water, and admixtures (e.g., superplasticizers, retarders) for strength and workability.
  - **Reinforcing Steel (Rebar):** Deformed steel bars embedded within the concrete to provide tensile strength and ductility, crucial for resisting bending moments and shear forces. Various diameters of rebar are visible, including larger diameter bars for columns and smaller bars for slabs.
- **Formwork Materials:**
  - **Timber/Plywood Sheathing:** Used for slab soffits and beam sides, supported by a system of joists and stringers.

3. Estimated Dimensions

Estimating precise dimensions from a partial image is challenging; however, an approximate assessment can be made:

- **Building Height (Under Construction):** At least 6-8 stories are visibly under various stages of construction or fully formed. Assuming an average floor-to-floor height of 3.2 to 3.8 meters (10.5 to 12.5 feet) for a typical multi-story building, the visible portion could range from approximately **20 to 30 meters (65 to 100 feet)**. The presence of a high-reaching tower crane suggests the final height will be significantly greater, potentially exceeding 15-20 stories (50+ meters).
- **Floor Plate Dimensions:** The building appears to have a substantial footprint. While the full extent is not visible, the visible section suggests floor dimensions of at least **30-50 meters (100-165 feet) in one dimension**, with the other dimension not fully discernible. The overall plan dimensions could easily encompass several hundred to over a thousand square meters per floor.
- **Column Dimensions:** Typical column sizes for such a structure might range from **400mm x 400mm (16" x 16") to 800mm x 800mm (32" x 32") or larger**, depending on location within the structure and accumulated loads.
- **Slab Thickness:** Appears to be a flat slab or beam-and-slab system. Typical slab thickness would likely be in the range of **180mm to 300mm (7" to 12")**, depending on span and loading.

4. Construction Techniques

The construction process visible employs standard reinforced concrete construction techniques:

- **Cast-in-Place Concrete Construction:** All primary structural elements (columns, beams, slabs) are being poured and cured on site.
- **Formwork Erection:** Timber and plywood formwork panels are meticulously erected and supported by adjustable steel/aluminum props (shoring) to create the desired shape of the concrete elements. This is a cyclical process, where formwork is stripped after concrete gains sufficient strength and then re-erected on the next level.
- **Reinforcement Installation:** Steel rebar cages are pre-assembled or assembled in-situ within the erected formwork. Proper spacing, cover, and lap lengths for splicing are critical and appear to be managed, with exposed rebar protruding from columns and slabs ready for continuity to the next level.
- **Concrete Placement:** Concrete is transported from a batching plant (off-site or on-site) and typically lifted by a **tower crane** using concrete buckets, then poured into the formwork. Vibrators are used to ensure proper consolidation and removal of air voids.
- **Curing:** Once poured, concrete is carefully cured (e.g., by covering, misting, or applying curing compounds) to achieve its specified strength and durability.
- **Staged Construction:** The project demonstrates typical multi-story construction phasing, with different floors simultaneously undergoing rebar installation, concrete pouring, and curing, allowing

5. Structural Integrity Observations

Based on the visible elements, the structural integrity appears to be competently managed during the construction phase:

- **Column & Beam Sizes:** The visible dimensions of the columns and beams appear proportionate to the scale of a multi-story building, suggesting they are designed to adequately carry the anticipated vertical loads.
- **Reinforcement Continuity:** Exposed rebar for column extensions and slab connections shows adequate lap lengths, which is critical for ensuring full transfer of forces between successive pours.
- **Formwork Support:** The presence of extensive shoring and props beneath newly poured slabs indicates proper support during the concrete curing period, preventing premature deflections or failures.
- **Concrete Finish:** While precise quality assessment is not possible from the image, the exposed concrete surfaces appear generally uniform and free from major defects such as excessive honeycombing or spalling at this stage.
- **Absence of Distress:** No obvious signs of structural distress such as significant cracks, deflections, or material failures are evident in the photograph.

6. Possible Engineering Challenges

Projects of this scale and type typically present several engineering challenges:

- **Lateral Load Resistance:** Designing the structure to effectively resist wind loads and potential seismic forces (depending on the geographical location) will require careful consideration of shear walls, moment frames, or a combination thereof.
- **Foundational System:** The design and construction of deep or shallow foundations (piles, rafts, or spread footings) suitable for the high vertical loads and varying soil conditions are critical. Differential settlement must be mitigated.
- **Concrete Quality Control:** Ensuring consistent concrete strength, slump, and durability across all pours, especially for high-rise elements, requires rigorous quality control procedures, including material testing and proper curing.
- **Construction Tolerances:** Maintaining tight tolerances for verticality of columns and alignment of slabs across multiple floors is crucial for structural performance and architectural finishes.
- **Progressive Collapse Mitigation:** Designing the structure to prevent disproportionate collapse in the event of a localized failure, as per modern building codes.
- **Logistics and Phasing:** Efficient scheduling of material delivery (especially concrete), crane operations, formwork cycling, and labor management across multiple floors.
- **Thermal Effects:** Managing the heat of hydration in large concrete pours, particularly for thick

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

1. Saves Documentation Time

The system significantly reduces the time required for manual structural documentation. Instead of writing detailed reports manually, engineers can upload an image and receive a structured description within seconds.

2. AI-Powered Automation

By integrating the Google Gemini Vision model, the application automatically analyses structural images and generates context-aware engineering descriptions. This minimizes human effort while maintaining professional quality.

3. Easy to Use

The web-based interface is simple and user-friendly. Users only need to upload an image to generate a structured report, making the system accessible to engineers, students, and construction professionals.

8.2 Disadvantages

1. Requires Internet Connection

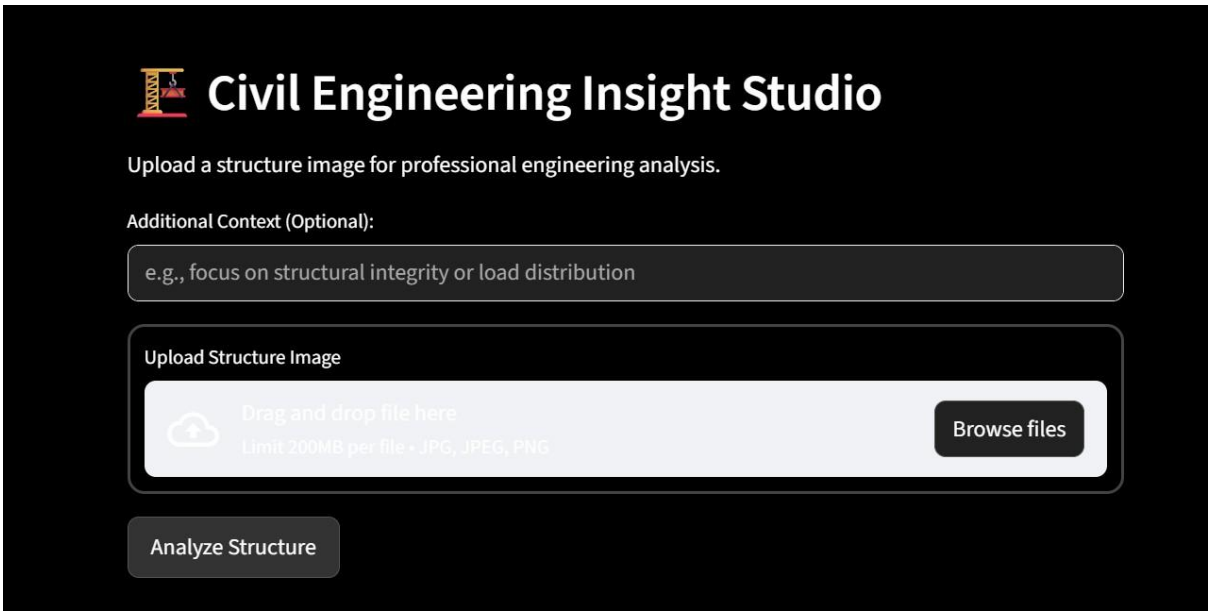
Since the system depends on the Google Gemini API for image processing, a stable internet connection is required.

2. Depends on API Availability

The application relies on external AI services. If the API service is unavailable, under maintenance, or exceeds usage limits, the system may not function properly.

3. Limited to Visible Analysis

The AI model analyses only what is visible in the image. It cannot assess internal structural strength, hidden defects, or load-bearing capacity without additional engineering data.



9. CONCLUSION

Civil Engineering Insight Studio is an innovative AI-powered web application designed to automate the process of structural image analysis and documentation.

The system successfully integrates artificial intelligence with civil engineering concepts to generate structured and professional descriptions of civil engineering structures.

By leveraging the advanced capabilities of the Google Gemini Vision model, the application can analyze uploaded images and identify structural components, materials, construction methods, and notable engineering features. This significantly reduces the manual effort required for documentation and improves reporting consistency.

The project demonstrates how AI can be effectively applied in the field of civil engineering to enhance productivity and efficiency.

It provides a practical solution for engineers, students, and professionals who require quick preliminary structural documentation.

Although the system depends on internet connectivity and external API services, it serves as a strong foundation for future

advancements in AI-assisted structural analysis. With further improvements such as damage detection.

BIM integration, and offline processing capabilities, the system can evolve into a comprehensive intelligent civil engineering assistant.

In conclusion, Civil Engineering Insight Studio successfully achieves its objective of automating structural description generation and represents a meaningful step toward integrating artificial intelligence into modern civil engineering practices.

10. FUTURE SCOPE

- Structural defect detection
- Crack identification
- PDF report generation
- Integration with construction tools

11. APPENDIX

GitHub Link: https://github.com/Manisairam143/PROJECT

Demo Video Link: **https://drive.google.com/file/d/1K-bMmGRGxmCRHfHr4gMNdzJZPSJl977s/view?usp=drive_link**